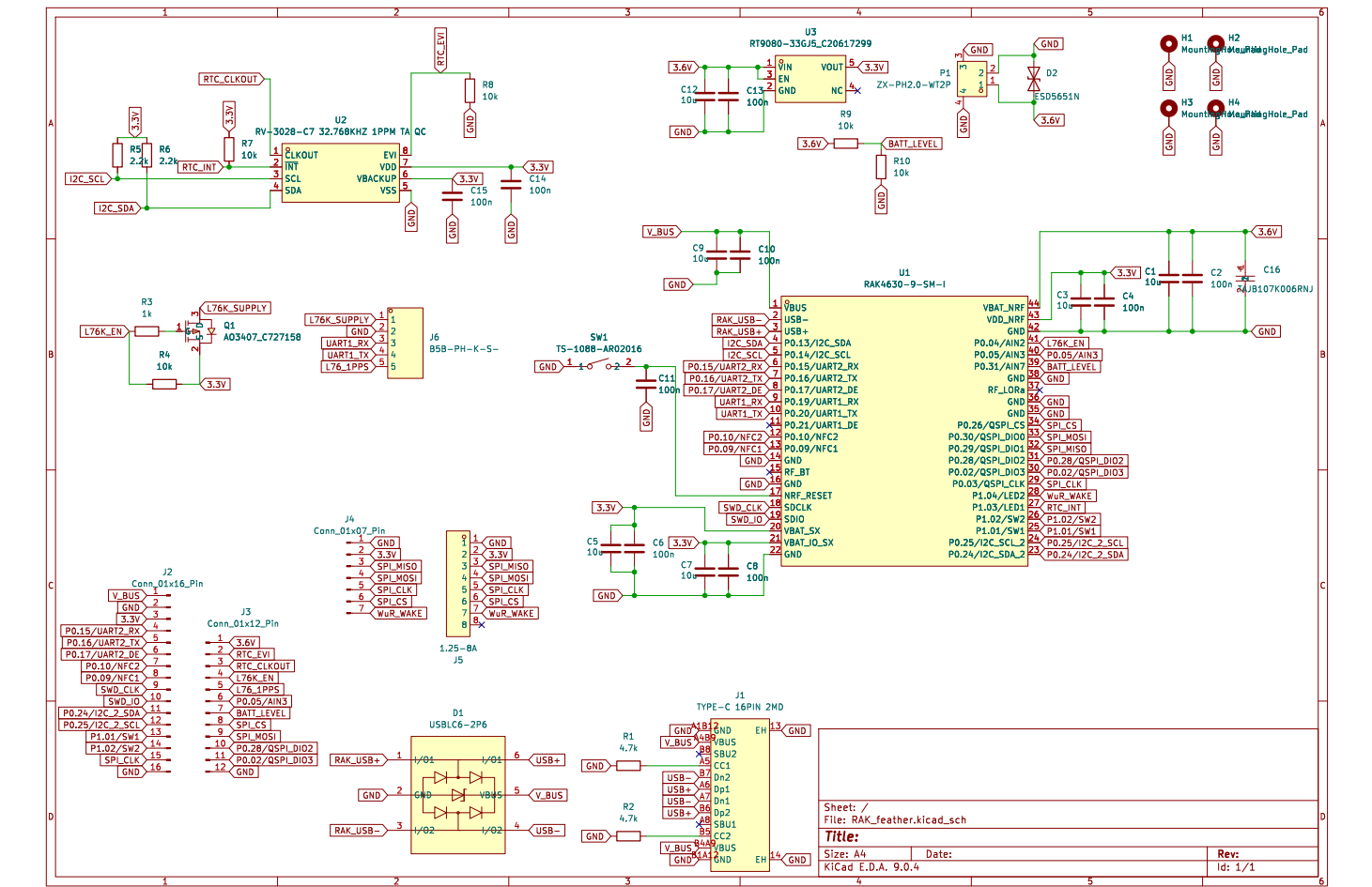
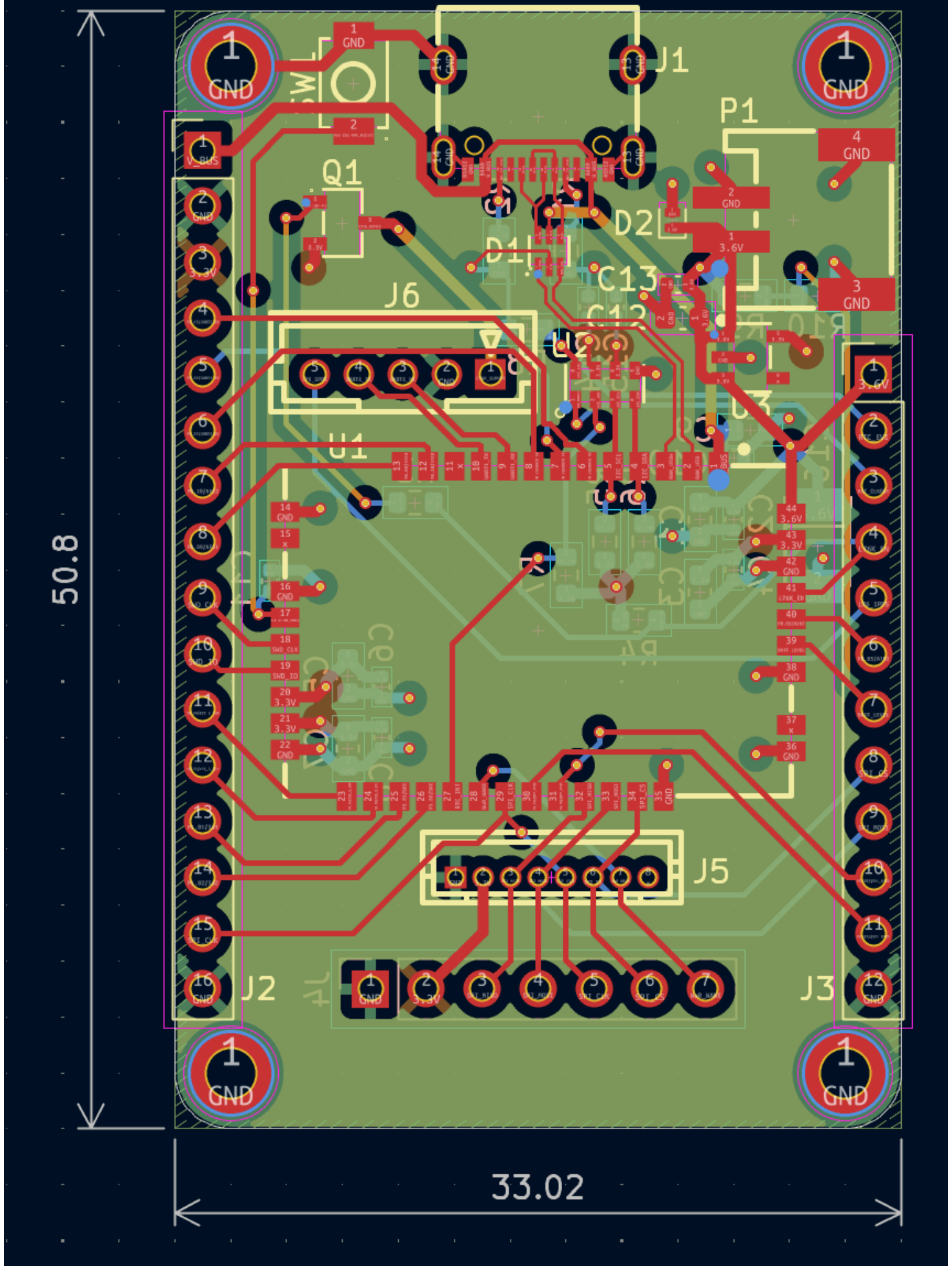


Schematic



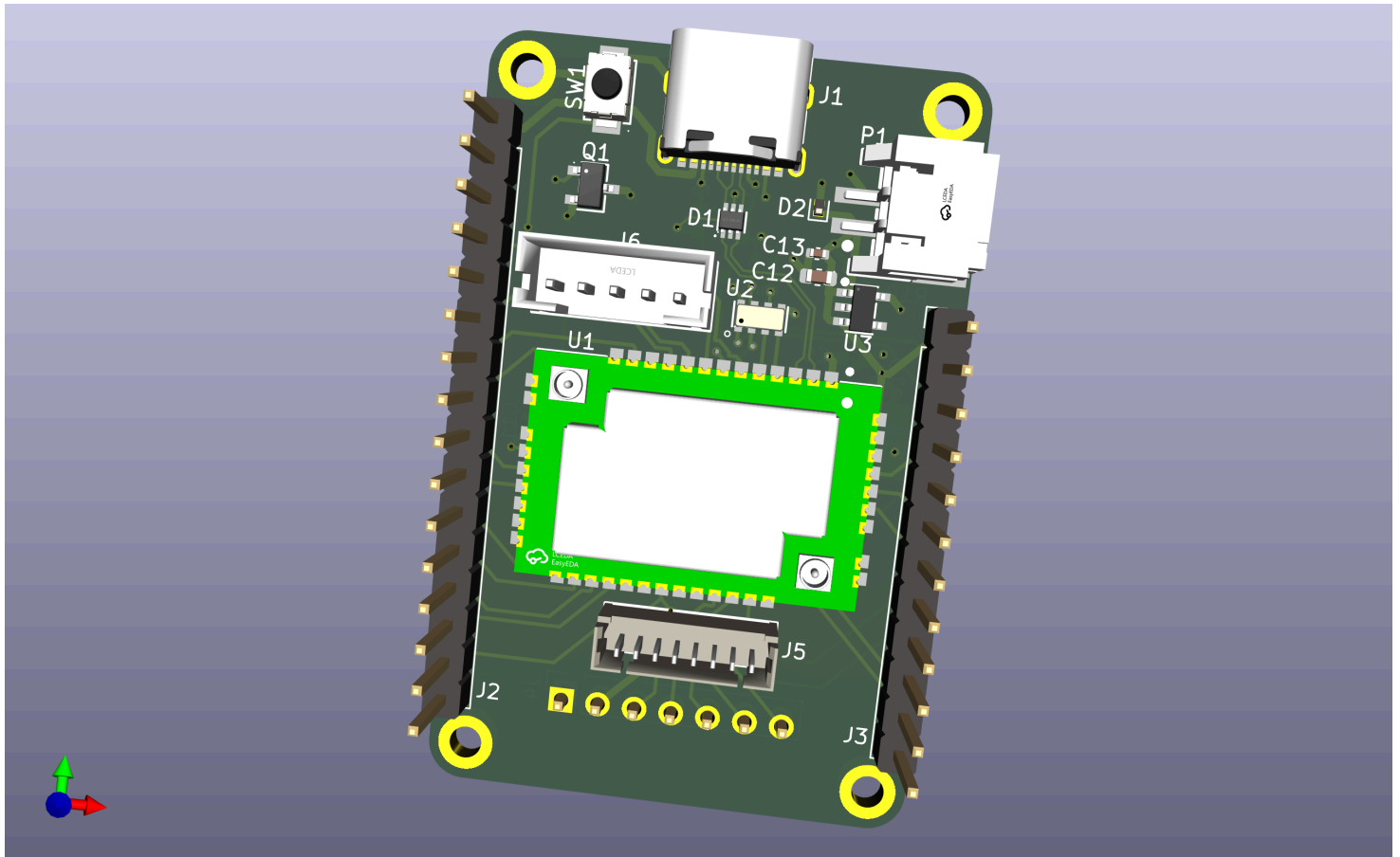
Layout, Dimensions, and Footprint

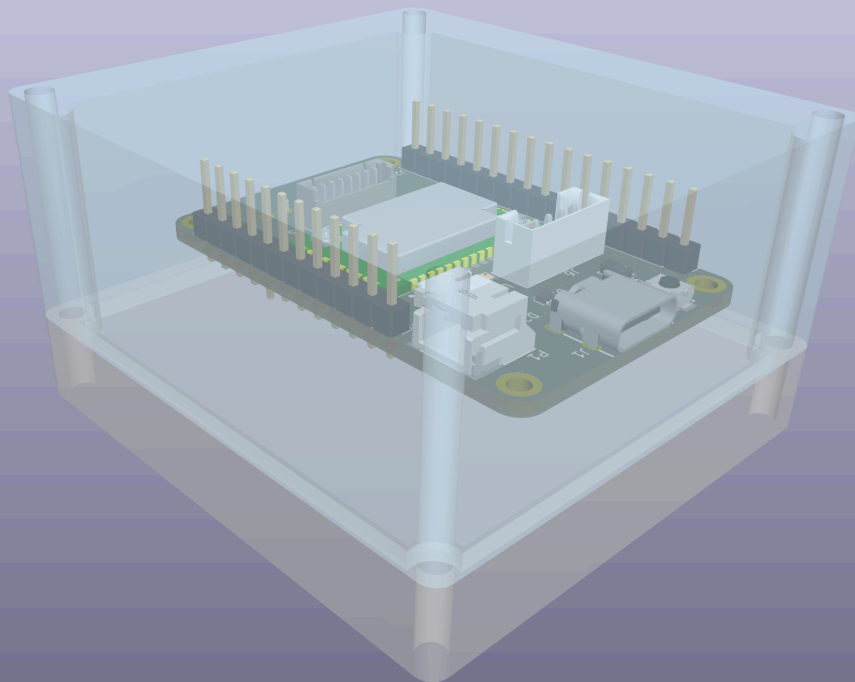
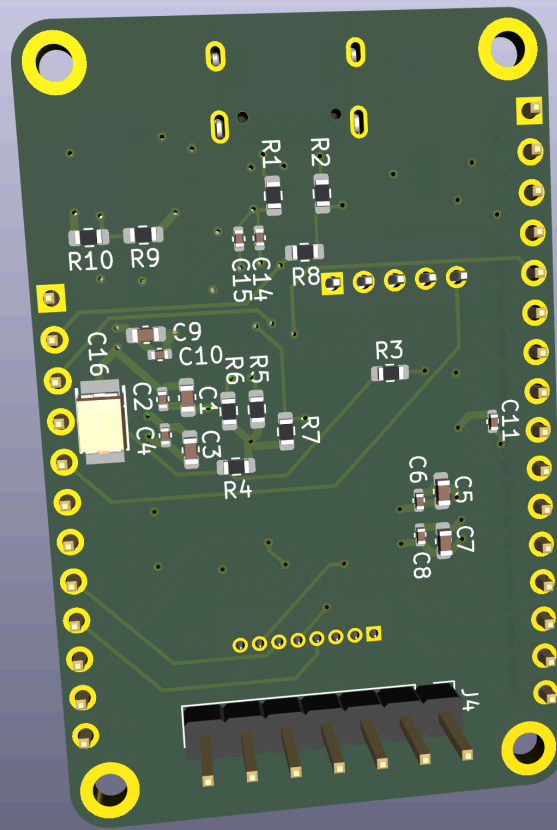


- 4-layer board -- JLC04161H-7628 stack up
- Wider version of typical Feather board (extended from mm to mm)
- Same height as typical Feather board
- 16-pin 2.54mm pitch male header (left)

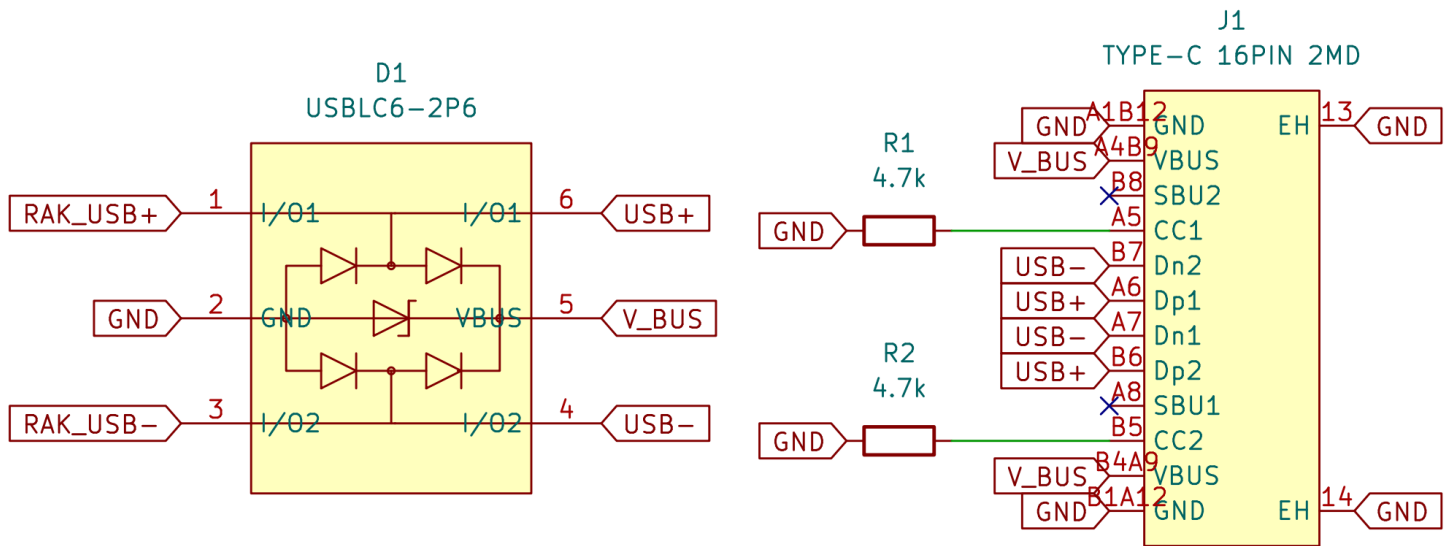
- 12-pin 2.54mm pitch male header (right)
 - M2 screw holes
 - PH 2.0 battery connector
 - USB-C connector
 - Reset pushbutton
 - 7-pin 2.54mm pitch male header on bottom for stacking with previously designed WuR board
-

3D Render

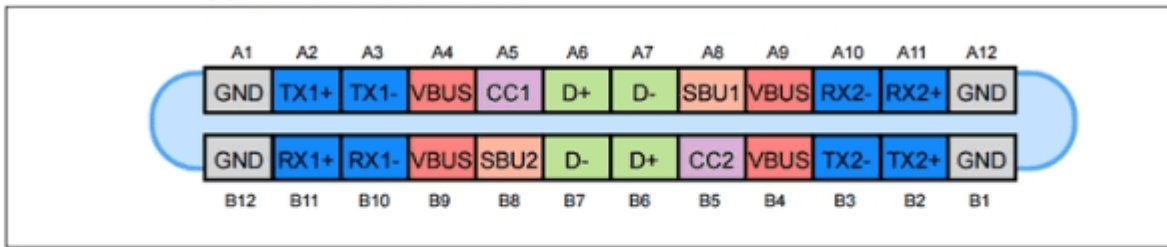




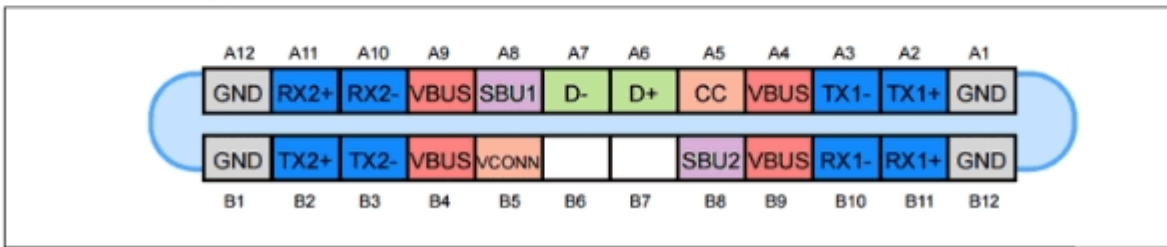
USB



USB TYPE-C RECEPTACLE

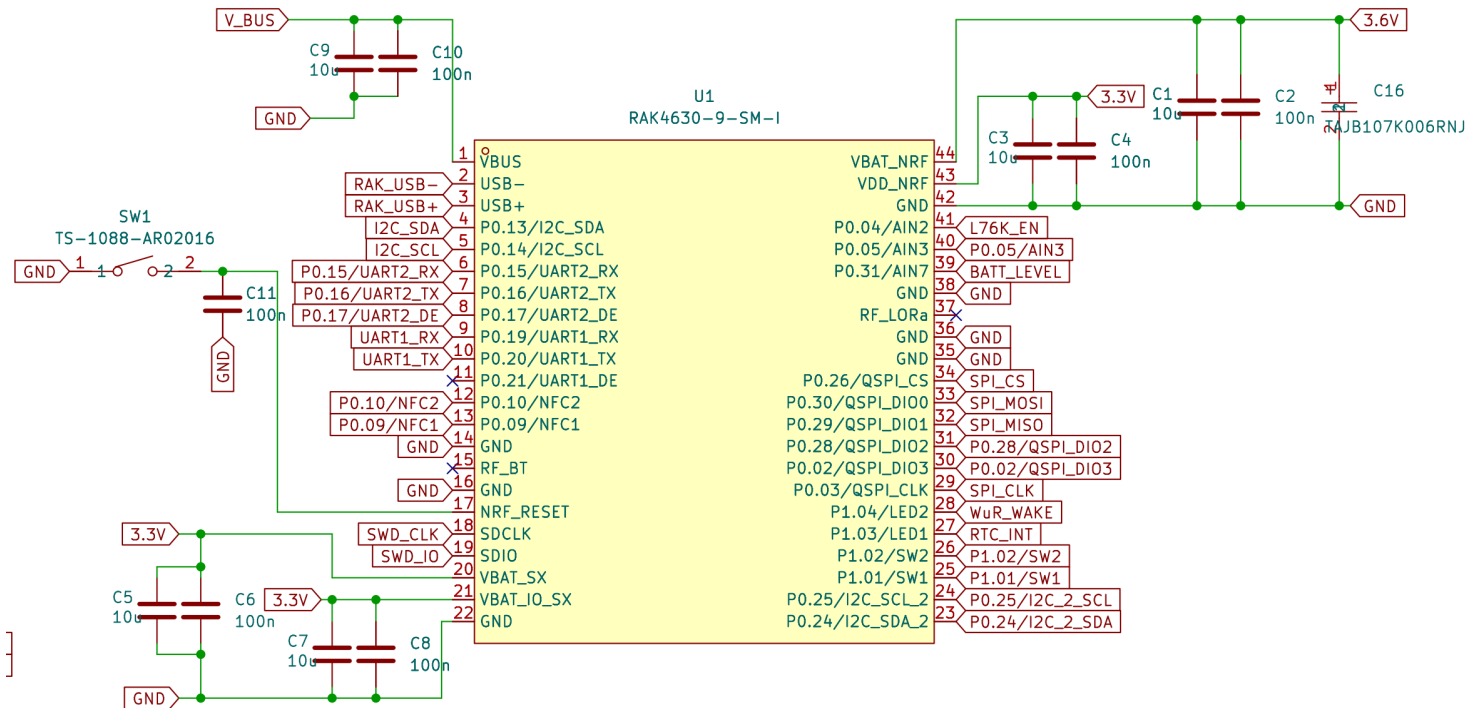


USB TYPE-C PLUG



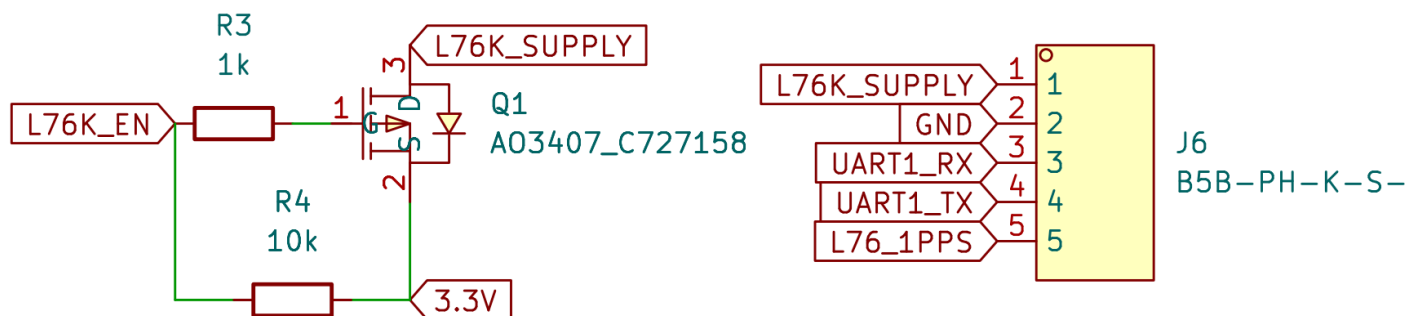
- 90Ω differential impedance on data lines
- 2nd differential pair to allow reversible connection
- CC pins unused, pulled to GND
- SBU pins unused, NC
- USBLC6-2P6 TVS diode for protection
- *check differential pair routing on PCB layout*

RAK 4630



- RF_LORA and RF_BT pins NC
- Reset pushbutton added
- 3.3V
 - V_DD_NRF
 - VBAT_SX
 - VBAT_IO_SX
- 3.6V (battery direct connection)
 - VBAT_NRF
- VBUS only powers USB peripherals on RAK, does not interfere with battery supply

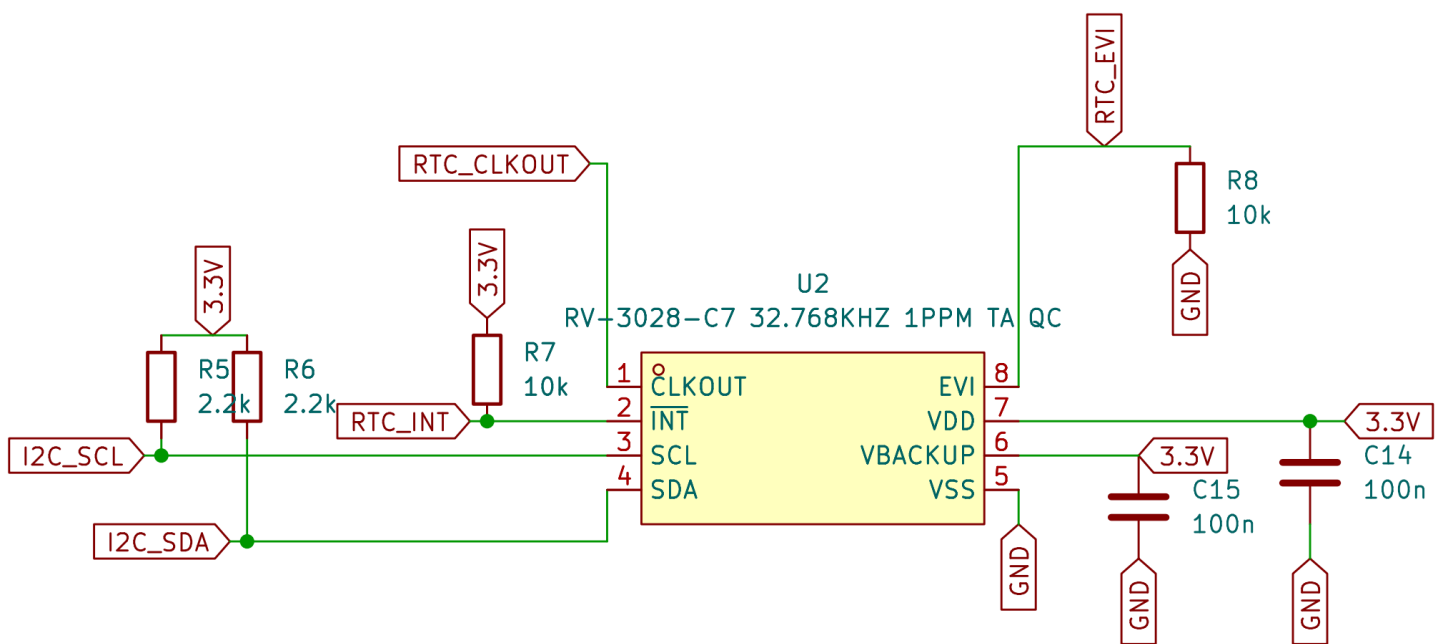
L76K GNSS Module





- Interfacing with [external board](#)
- Using 5-pin 2mm pitch wire-to-board connector
- UART communication (2 pins) with RAK
- Transistor power control implemented with AO3407 P-MOSFET

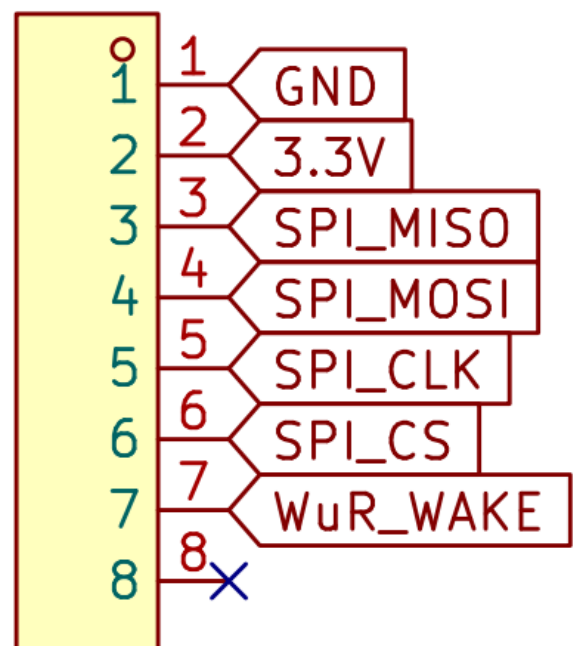
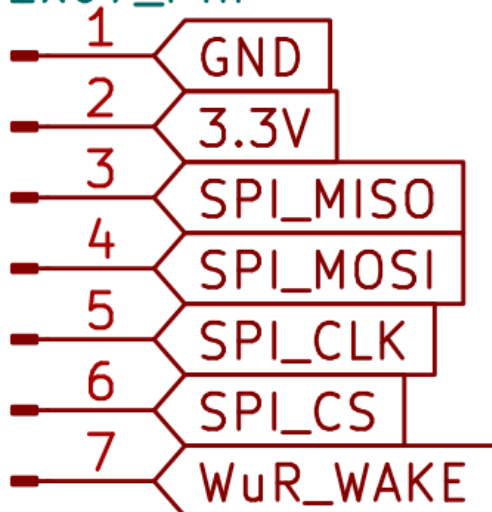
RTC



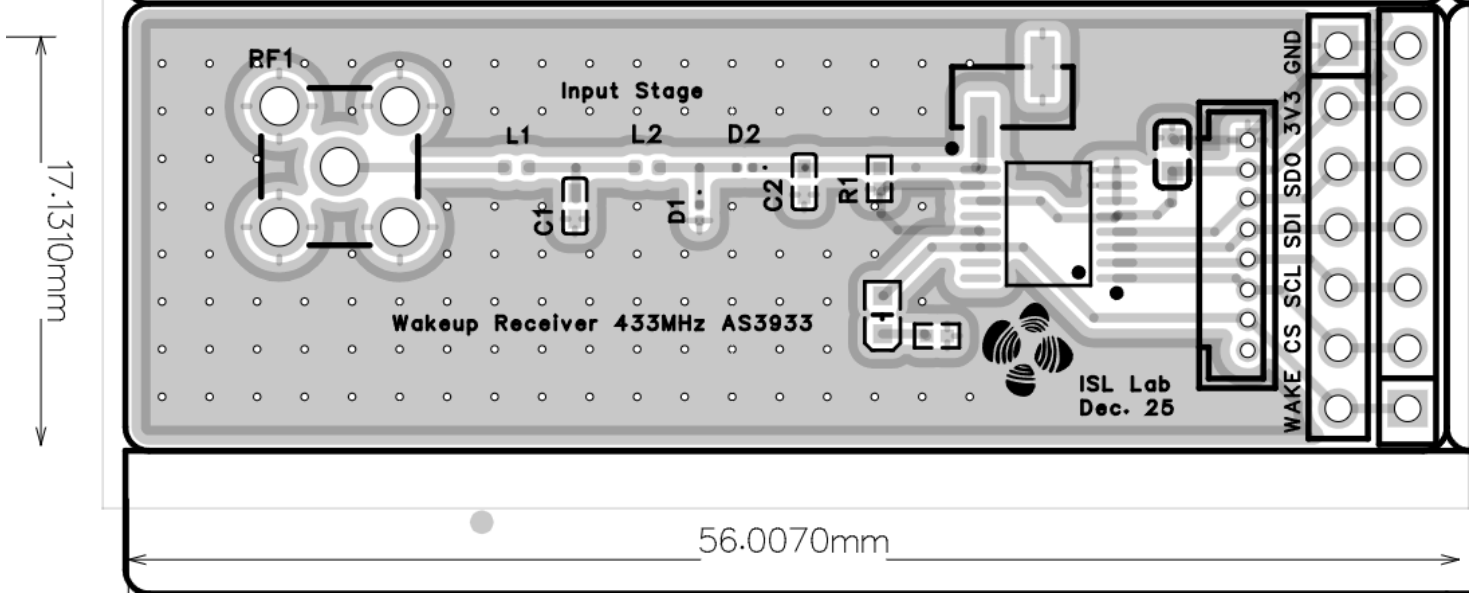
- Followed "NON-RECHARGEABLE BACKUP SOURCE / EVENT INPUT USED (ACTIVE HIGH)" schematic in the application manual
- Backup supply voltage same as main supply (3.3V)
- I2C communication (2 pins with pull-up resistors) with RAK
- RTC_INT output (active low) connected to RAK GPIO pin

WuR

J4
Conn_01x07_Pin

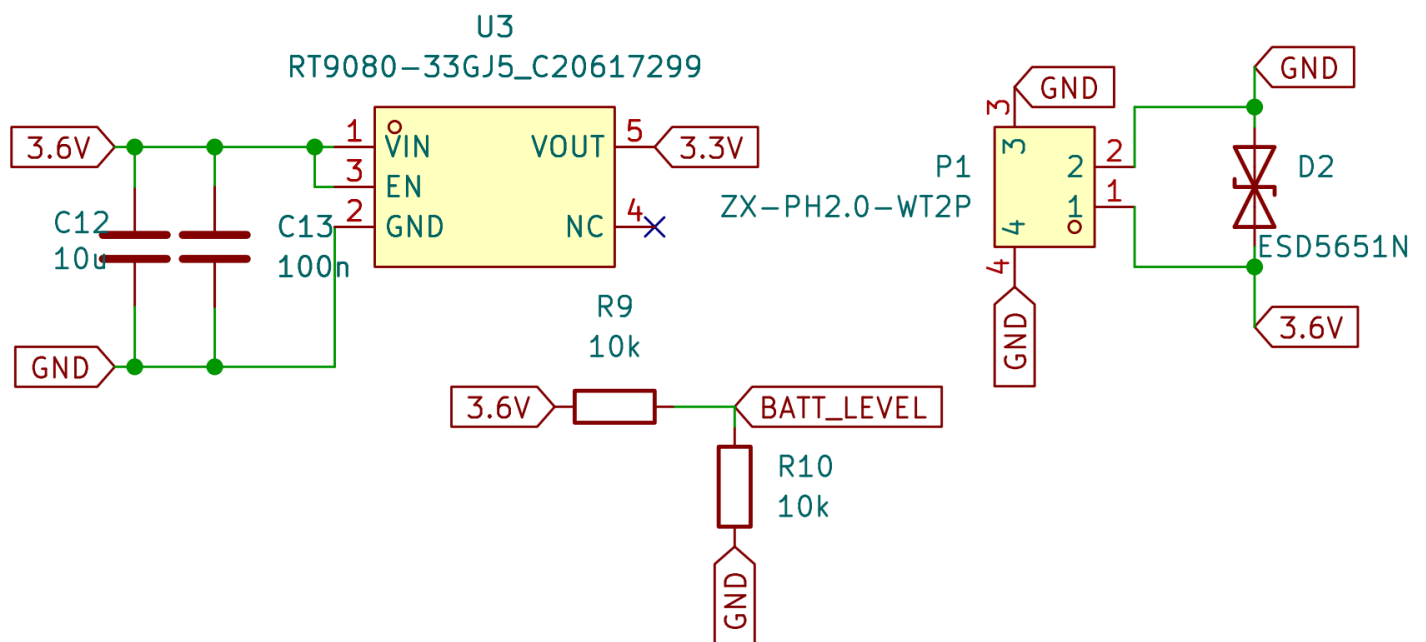


1.25-8A
J5



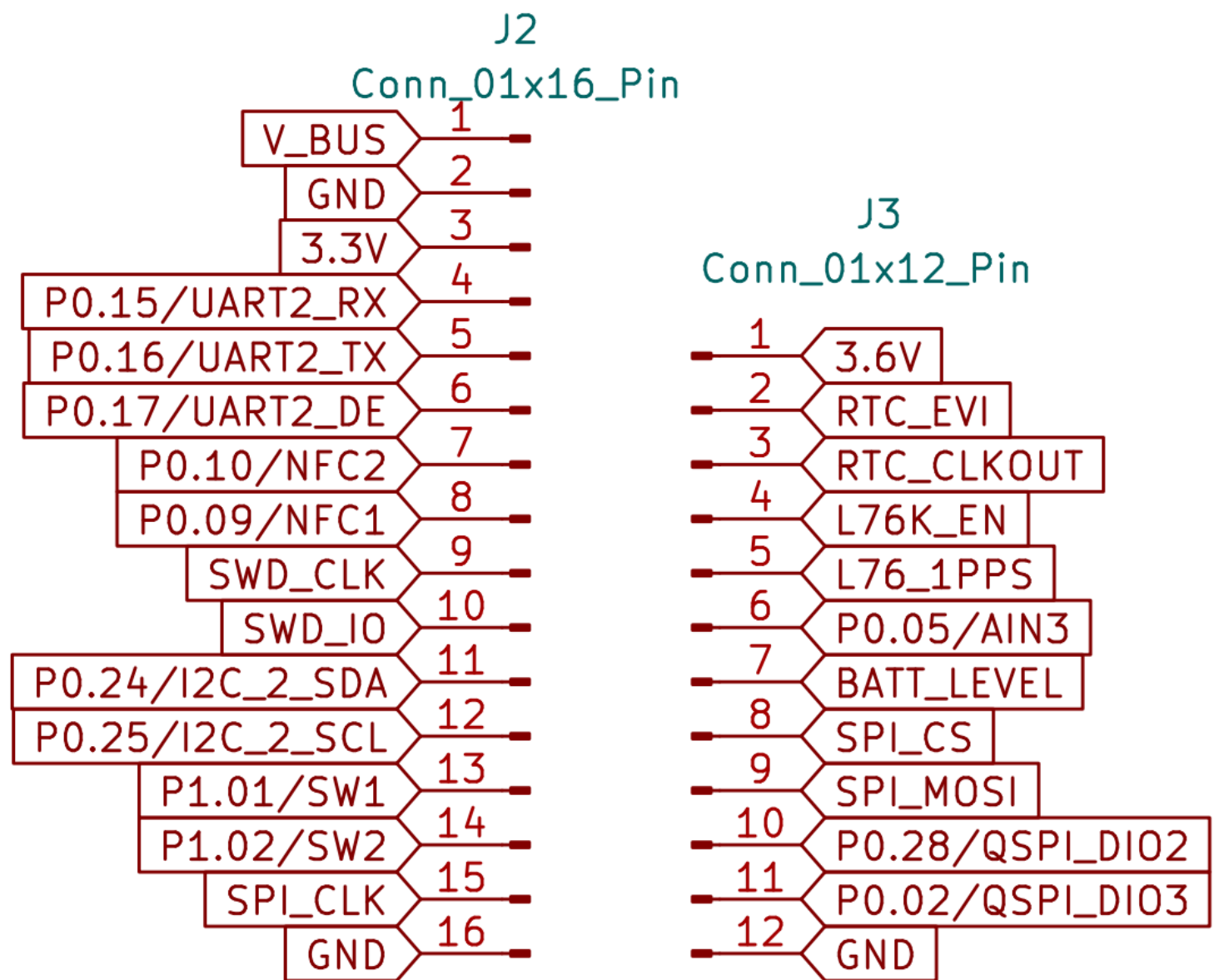
- Using existing board design
- RAK board will stack on top and connect with 7-pin header
- SPI communication (4 pins) with RAK
- WuR_WAKE signal output connected to RAK GPIO pin

Power Supply



- 3.6V battery as the main supply
- PH 2.0 connector
- ESD5651 TVS diode for protection
- Assuming battery is always present
 - Battery supply is converted to 3.3V
 - USB 5V does not power modules, only USB peripheral of nRF
- Voltage divider for battery level analog signal
 - Same value resistors used ($10k\Omega$)
- BATT_LEVEL connected to RAK analog input pin

Header Pins



- Priority for unused pins on RAK, GNSS module, and RTC

Passive Components

- Resistors
 - 0603 footprint for all
- Capacitors
 - 10 μF (C2959727 JLC part number)
 - 0603 footprint
 - 10V rating
 - X7R
 - 10% tolerance
 - 100nF (C1525 JLC part number)
 - 0402 footprint
 - 16V rating

- X7R
 - 10% tolerance
 - 100 μF tantalum capacitor (C16133 JLC part number)
 - CASE-B-3528 footprint
 - 6.3V rating
 - 10% tolerance
-